Description of nearest neighbor search method

In order to determine the location of mobile devices using a wireless network, it is first necessary to create a map of the radio signal strengths of the area of the building where these devices will be monitored.

When creating a map of radio signal strengths, the monitored area of the building is divided into a grid, at the intersection points of which the radio signal strengths (hereinafter RSS) are measured. Suppose there are N visible access points (hereafter AP) in the map area. Then, at each j-th point of intersection of the grid lines, the signal strengths up to the visible APs can be measured and the RSS vector $M_j = <\mu_{j1}, \mu_{j2}, \ldots, \mu_{jN}$., where μ_{ji} is the RSS to the corresponding AP_i.

Each database entry corresponds to the intersection point $x_j;y_j$ of the grid lines and the RSS vector M_j.

When searching for a user's location on a wireless network, the signal strengths from the mobile device to all visible APs are first measured. The RSS vector $R=<r_1, r_2, \ldots, r_N>$ is formed, where r_i is the RSS to each visible AP_i. By comparing R and M_j, it is possible to find a j at which the value of the difference between R and M_j is the smallest. The smallest values of R and M_j indicate that the geographical locations corresponding to them are the least distant from each other compared to other locations of the RSS grid. Therefore, we can conclude that the object is at location ($x_j; y_j$).

In the nearest neighbor search method, the difference between two vectors (let's say in this case R and M) is calculated according to the formula (1). It is called the Euclidean distance. Let D_j be the difference between the R and M vectors. D_j is then calculated as follows:

$$D_j = \sqrt{\sum_{i=1}^{N} (r_i - \mu_{j,i})^2} \qquad (1)$$

The value of j at which D_j is the smallest will be a point on the RSS map grid indicating the location of a mobile device using wireless network resources.

Example

Let's say we have a radio signal strength map, where radio signals are measured at 9 points (see Figure 1). Here j is the measurement sequence number.

At each point, the signal strengths were measured for up to 3 access stations (see Table 1). The signal strengths from the mobile device to be located to the AP are described by the vector R, where $R=(24;46;40)$.

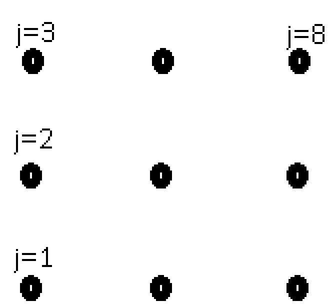

Figure 1. RSS map measurement points

Table 1. Signal strengths at each measurement point *j* of the RSS map.

j – measurement sequence number	Contractual coordinates (x;y)	RSS iki AP_1	RSS iki AP_2	RSS iki AP_3
1	$x_0; y_0$	50	30	50
2	$x_0;y_1$	40	40	50
3	$x_0;y_2$	30	50	50
4	$x_1;y_0$	40	30	40
5	$x_1;y_1$	30	40	40
6	$x_1;y_2$	20	50	40
7	$x_2;y_0$	30	30	50
8	$x_2;y_1$	20	40	50
9	$x_2;y_2$	10	50	50

Table 2. Euclidean distance D_j for each point *j* of the RSS map.

j – measurement sequence number	Contractual coordinates (x;y)	D_j
1	$x_0; y_0$	32,12476
2	$x_0;y_1$	19,79899
3	$x_0;y_2$	12,32883
4	$x_1;y_0$	22,62742
5	$x_1;y_1$	8,485281
6	$x_1;y_2$	5,656854
7	$x_2;y_0$	19,79899
8	$x_2;y_1$	12,32883
9	$x_2;y_2$	17,66352

To determine the location of the mobile device in the given area, we calculate the Euclidean distance D_j for each measurement point *j* of the RSS map according to the expression (*1*) (calculation results are presented in Table 2). The lowest value was obtained when $j=6$. The coordinates where this RSS map measurement was taken will be the location of the mobile object (see Figure 2).

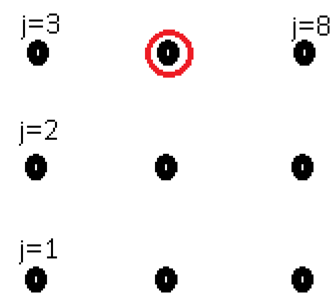

Figure 2. The location of the mobile device is determined when *R=(24;46;40).*

Used literature

Q. Tran, J. Wirawan Tantra, C. Heng Foh, A. Tan, K.Yow, D. Qiu. Wireless Indoor Positioning System with Enhanced Nearest Neighbors in Signal Space Algorithm. Nanyang Technological University, Singapore, Concordia University, Canada, 2006 IEEE.

The original source is attached to the laboratory work material as additional literature.